

COURSE CODE: DSA0601

COURSE NAME: DATA HANDLING AND VISUALIZATION

Particulars	Details
Assignment Title	From Raw Data to Decision-Ready Dashboards: Analyzing Retail Sales Patterns, Evaluating Visualization Effectiveness, and Creating an Interactive Customer Analytics Dashboard
Course Outcome(s) – CO	CO3: Analyze data patterns, trends, and associations using various visualization techniques.CO4: Evaluate visualization designs based on perception, cognition, and effectiveness principles.CO5: Create effective and interactive visualizations by integrating data wrangling and advanced visualization tools.
Bloom's Taxonomy Level	L4 – AnalyseL5 – EvaluateL6 – Create
SDG Mapping (SDG 1–17)	SDG 8 – Decent Work and Economic GrowthSDG 12 – Responsible Consumption and Production

Dataset: food_delivery_customer_dataset

1. Problem Statement and Problem Formulation

1.1 Problem Statement

Food delivery platforms generate large amounts of customer transaction data containing information such as restaurants, cuisines, cities, order values, discounts, delivery time, customer ratings, payment methods, and promotions. Raw data in tabular form makes it difficult to identify important patterns and trends for effective decision-making.

This project focuses on analyzing a food delivery customer dataset and developing meaningful visualizations and an interactive dashboard. The system processes the dataset, identifies customer ordering patterns, analyzes category and city-level performance, studies relationships between numerical variables, and presents the results through effective visualizations.

The implementation demonstrates important Data Handling and Visualization concepts, including data cleaning, data wrangling, trend analysis, correlation analysis, visualization evaluation, and interactive dashboard design.

1.2 Problem Formulation

The input consists of food delivery customer transaction records containing order details, restaurant information, cuisine, city, meal time, order value, discount percentage, delivery time, customer rating, payment method, and promotion details.

The data is processed through the following stages:

Raw Dataset → Data Cleaning → Data Wrangling → Exploratory Analysis → Visualization
→ Pattern Analysis → Interactive Dashboard

The dataset is first checked for missing values and duplicate records. Date values are converted into a suitable format for time-based analysis. The processed data is then used to identify trends, category-level performance, city-wise performance, and associations between variables.

The system should also provide interactive filtering and clear visual representations to support faster and more accurate decision-making.

2. Objective and Expected Outcomes

2.1 Objectives

The main objective of this project is to analyze food delivery customer data and develop an effective interactive analytics dashboard using Data Handling and Visualization concepts.

The objectives are:

1. To clean and preprocess the food delivery dataset.
2. To identify trends and patterns in customer orders.
3. To analyze cuisine-wise and city-wise performance.
4. To study meal-time ordering patterns.
5. To analyze relationships between order value, discount, delivery time, and customer rating.
6. To evaluate visualization designs using perception and cognition principles.
7. To create effective and interactive visualizations.
8. To develop a dashboard with suitable filters.
9. To generate meaningful insights for decision-making.

2.2 Expected Outcomes

The expected outcome is a working interactive customer analytics dashboard that can analyze food delivery transaction data and present important business insights.

The project should provide evidence showing that the data has been properly processed and that visualizations can reveal trends, comparisons, and associations. The interactive dashboard should allow users to explore the data using filters such as city, cuisine, and date range.

3. Requirements, Constraints and Assumptions

3.1 Requirements

The system should be able to:

- Load and process the food delivery dataset.
- Check for missing values and duplicate records.

- Convert date values into suitable format.
- Analyze order trends over time.
- Compare cuisine-wise and city-wise performance.
- Analyze meal-time ordering patterns.
- Generate correlation analysis.
- Create suitable visualizations.
- Evaluate visualization effectiveness.
- Provide interactive dashboard filters.

3.2 Constraints

The implementation is limited to the variables available in the provided dataset. The dataset contains only 200 records, so the analysis represents the available sample rather than a complete real-world food delivery system.

The project focuses on data analysis and visualization and does not perform real-time delivery tracking or prediction. The effectiveness of the analysis depends on the quality and size of the available dataset.

3.3 Assumptions

The following assumptions are made:

- Each Order_ID represents a unique food delivery transaction.
- Order_Value represents the total value of an order.
- Customer_Rating represents customer satisfaction.
- Delivery_Time_Min represents the delivery duration.
- Discount_Percent represents the discount applied to an order.
- The dataset records are assumed to contain valid food delivery information.
- The analysis is performed using historical transaction data.

4. Application of Relevant Course Knowledge / Concepts

The project applies the main Data Handling and Visualization concepts required for transforming raw food delivery data into meaningful insights.

- Data Cleaning: Checks missing values and duplicate records.
- Data Wrangling: Converts and organizes the dataset into an analysis-ready format.
- Trend Analysis: Identifies changes in order values over time.
- Comparative Visualization: Compares cuisines, cities, and meal-time performance.

- Correlation Analysis: Identifies relationships between numerical variables.
- Visualization Evaluation: Evaluates charts using perception and cognition principles.
- Interactive Visualization: Allows users to explore data using filters and interactions.
- Dashboard Design: Combines multiple visualizations into a single decision-support interface.

4. Design / Proposed Solution / Methodology

4.1 Proposed Solution

The proposed system is an interactive Food Delivery Customer Analytics Dashboard that transforms raw transaction data into meaningful visual insights.

The dashboard analyzes customer ordering patterns based on cuisine, city, meal time, promotions, order values, delivery time, and customer ratings. Different visualizations are used to present trends and comparisons clearly.

4.2 Methodology

The data analysis process consists of the following steps:

1. Load the food delivery customer dataset.
2. Inspect the dataset structure and data types.
3. Check for missing values.
4. Check for duplicate records.
5. Convert Order_Date into datetime format.
6. Create time-related features where required.
7. Perform exploratory data analysis.
8. Generate trend and comparison visualizations.
9. Perform correlation analysis.
10. Evaluate visualization effectiveness.
11. Design improved visualizations.
12. Create an interactive dashboard.
13. Test dashboard filters and interactions.
14. Record and evaluate the results.

5.3 System Flow

The proposed system follows this sequence:

Food Delivery Dataset → Data Cleaning → Data Wrangling → Data Analysis → Visualization → Evaluation → Interactive Dashboard

The design provides a clear separation between data preprocessing, analysis, visualization, and dashboard development. The generated visualizations help users understand the patterns present in the dataset.

5. Algorithm / Pseudocode / Flowchart

5.1 Algorithm

Input: Food Delivery Customer Dataset

Output: Interactive Analytics Dashboard and Data Insights

1. Start the system.
2. Load the CSV dataset.
3. Check the dataset structure.
4. Identify missing values.
5. Identify duplicate records.
6. Convert Order_Date into datetime format.
7. Analyze order value trends.
8. Analyze cuisine-wise performance.
9. Analyze city-wise performance.
10. Analyze meal-time patterns.
11. Calculate correlations between numerical variables.
12. Generate visualizations.
13. Evaluate visualization designs.
14. Create an interactive dashboard.
15. Apply filters and interactions.
16. Display the results.
17. Stop.

5.2 Pseudocode

BEGIN

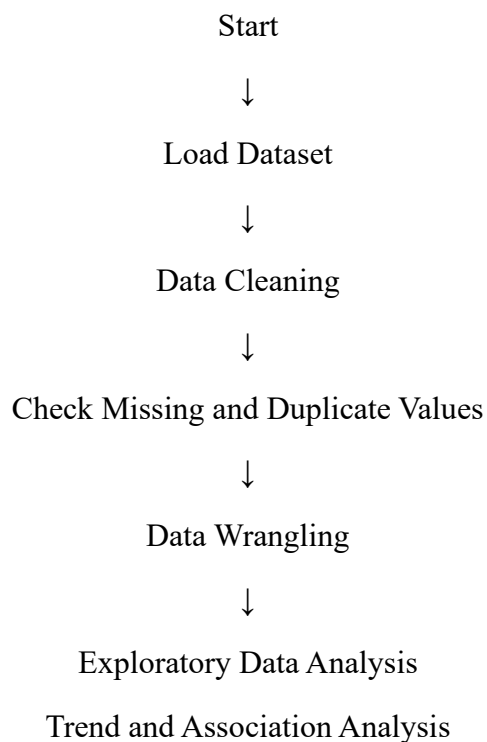
Load food delivery dataset

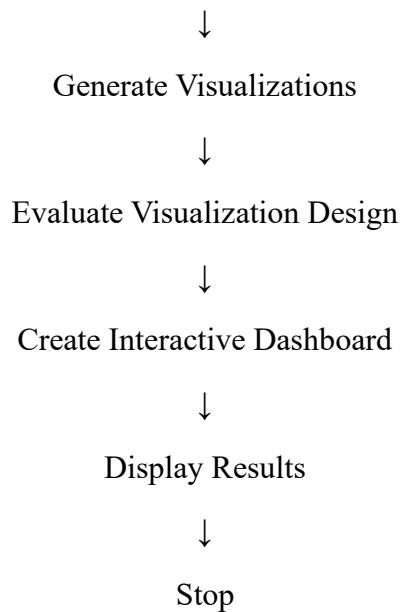
Check missing values

Check duplicate records
Convert Order_Date to datetime format
Create required analysis features
Calculate total orders and order value
Analyze monthly trends
Analyze cuisine-wise performance
Analyze city-wise performance
Analyze meal-time patterns
Calculate correlation between numerical variables
Generate visualizations
Evaluate visualization effectiveness
Create interactive dashboard
Apply filters
Display insights
END

5.3 Flowchart

The flowchart for the proposed data analytics system follows this sequence:





6. Implementation / Source Code and Environment / Tools Used

6.1 Implementation

The project is implemented in Python using data analysis and visualization libraries. The food delivery dataset is processed using Pandas for data handling and wrangling.

The dataset is checked for missing values and duplicate records before analysis. The `Order_Date` column is converted into datetime format to support time-based analysis. Different visualizations are generated to study trends, category performance, and relationships between numerical variables.

The interactive dashboard is developed using Plotly and Dash to provide filters and interactive charts.

6.2 Implementation Flow

The Python implementation follows:

CSV Dataset → Data Cleaning → Data Wrangling → Analysis → Visualization → Interactive Dashboard

The generated outputs from each stage are used to verify that the data processing and visualization process is functioning correctly.

6.3 Source Code

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

df = pd.read_csv("food_delivery_customer_dataset_200.csv")

print(df.head())
```

```
print(df.info())

print("\nMissing Values:")

print(df.isnull().sum())

print("\nDuplicate Records:")

print(df.duplicated().sum())

df["Order_Date"] = pd.to_datetime(df["Order_Date"])

df["Year"] = df["Order_Date"].dt.year

df["Month"] = df["Order_Date"].dt.month

print(df.describe())

monthly_sales = df.groupby(

    df["Order_Date"].dt.to_period("M")

)["Order_Value"].sum()

monthly_sales.plot(

    kind="line",

    marker="o",

    figsize=(10, 5)

)

plt.title("Monthly Order Value Trend")

plt.xlabel("Month")

plt.ylabel("Total Order Value")

plt.xticks(rotation=45)

plt.tight_layout()

plt.show()

cuisine_sales = df.groupby(

    "Cuisine"

)["Order_Value"].sum().sort_values()

cuisine_sales.plot(

    kind="barh",

    figsize=(8, 5)
```



```
)  
plt.title("Cuisine-wise Order Value")  
plt.xlabel("Total Order Value")  
plt.ylabel("Cuisine")  
plt.tight_layout()  
plt.show()  
  
city_sales = df.groupby(  
    "City"  
)["Order_Value"].sum().sort_values()  
city_sales.plot(  
    kind="bar",  
    figsize=(8, 5)  
)  
  
plt.title("City-wise Performance")  
plt.xlabel("City")  
plt.ylabel("Total Order Value")  
plt.xticks(rotation=45)  
plt.tight_layout()  
plt.show()  
  
meal_sales = df.groupby(  
    "Meal_Time"  
)["Order_Value"].sum()  
meal_sales.plot(  
    kind="bar",  
    figsize=(8, 5)  
)  
  
plt.title("Order Value by Meal Time")  
plt.xlabel("Meal Time")  
plt.ylabel("Order Value")
```

```

plt.tight_layout()
plt.show()
numeric_columns = [
    "Order_Value",
    "Discount_Percent",
    "Delivery_Time_Min",
    "Customer_Rating"
]
correlation = df[numeric_columns].corr()
plt.figure(figsize=(8, 6))
sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)
plt.title("Correlation Heatmap")
plt.show()

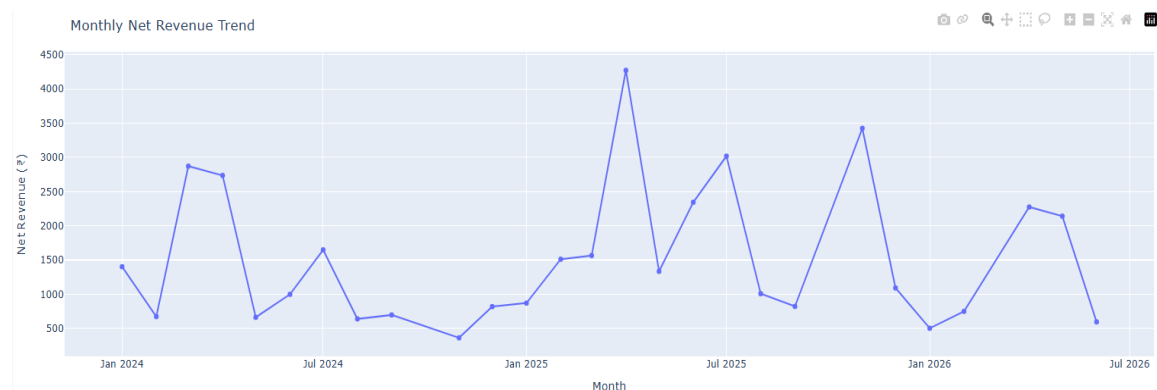
```

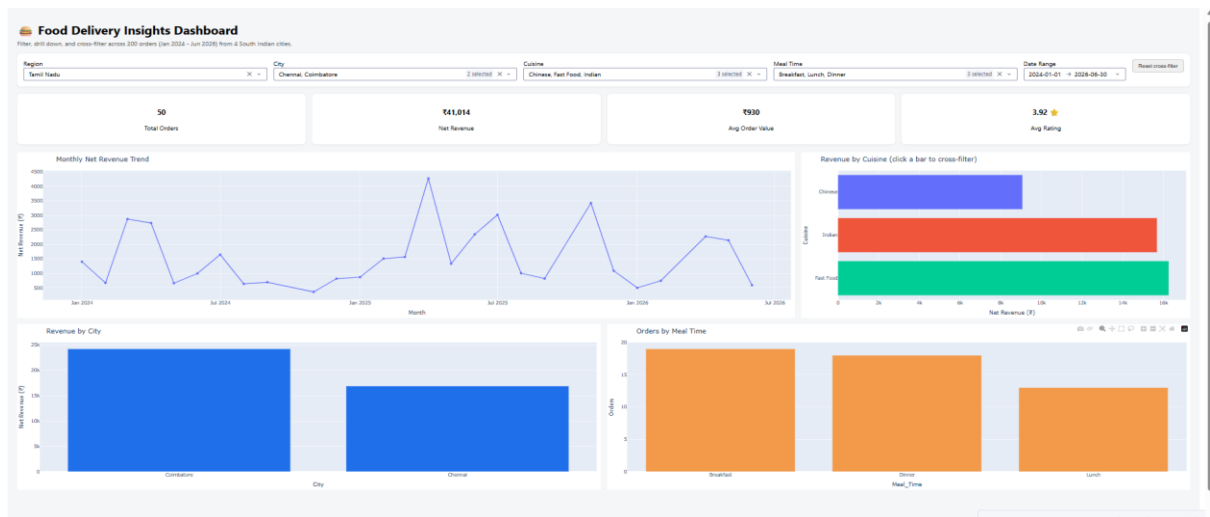
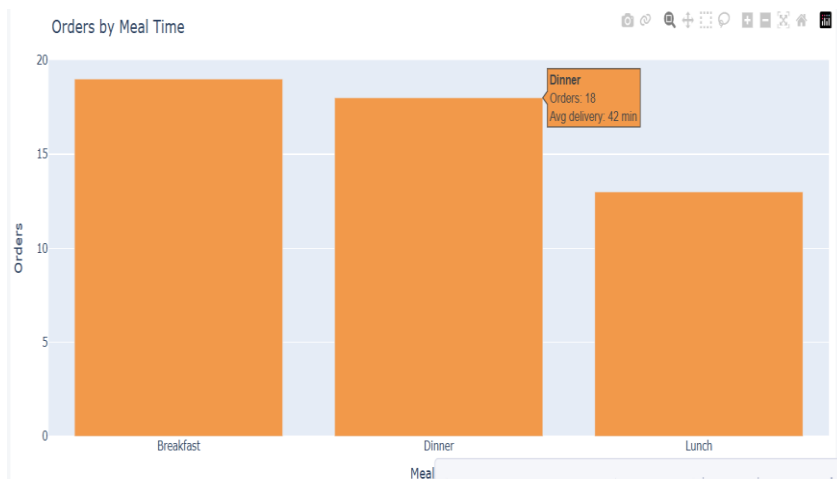
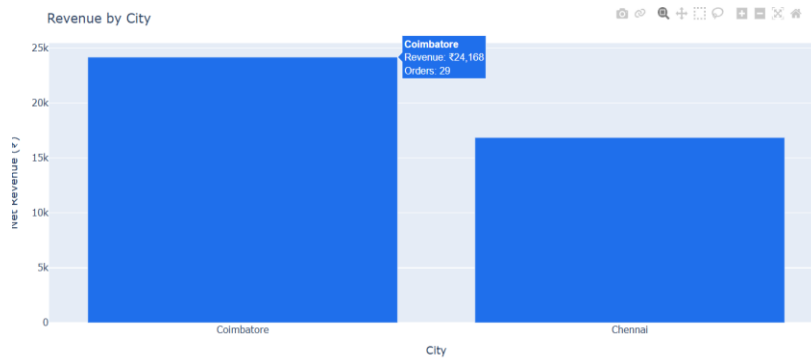
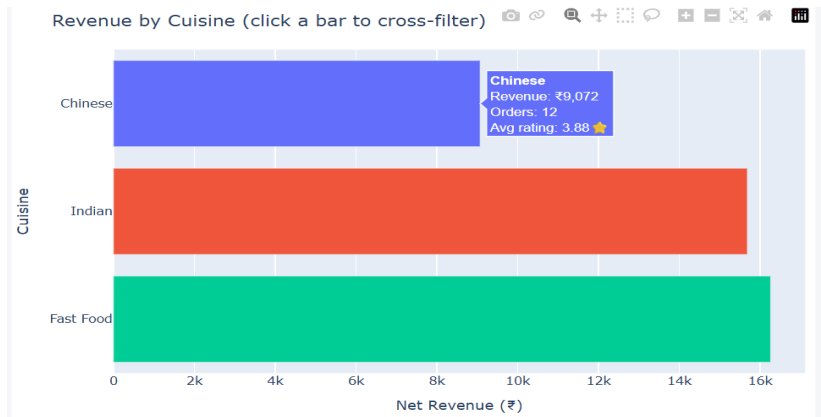
7. Execution Screenshots / Outputs

The implemented Python data analysis program was executed using the food delivery customer dataset.

The execution produced outputs for the major analysis stages, including:

The program successfully processed the provided dataset without data quality errors.





The execution screenshots provide evidence of successful data processing and visualization generation.

8. Results and Validation

The implemented system was validated using the provided food delivery customer dataset. The dataset was successfully processed through data cleaning, data wrangling, exploratory analysis, and visualization stages.

8.1 Experimental Observation

The analysis shows that the food delivery dataset can be effectively transformed into meaningful visual representations.

The trend analysis helps identify changes in order values over time. Cuisine-wise and city-wise visualizations allow comparison between different categories and locations. The correlation heatmap provides information about relationships between numerical variables such as order value, discount percentage, delivery time, and customer rating.

8.2 Validation of Data Analysis Stages

The execution output provides evidence that the major data processing stages were successfully completed. The dataset was loaded correctly, no missing or duplicate records were identified, and the date column was processed for trend analysis.

The successful generation of charts confirms that the processed dataset can be used to identify patterns and associations.

9. Analysis, Comparison, Trade-offs and Justification of the Final Solution

9.1 Analysis

The implemented system successfully processes food delivery customer data and converts it into meaningful visual insights.

The analysis focuses on important variables such as cuisine, city, meal time, order value, discount percentage, delivery time, customer rating, payment method, and promotion.

Trend charts help identify changes over time, while bar charts make category comparisons easier. Correlation analysis provides an overview of relationships between numerical variables.

9.2 Comparison

The following comparison shows the difference between traditional data presentation and the proposed interactive dashboard.

Parameter	Traditional Data Table	Interactive Dashboard
Trend Identification	Difficult	Easy

Parameter	Traditional Data Table	Interactive Dashboard
Data Exploration	Limited	Interactive
Category Comparison	Time-consuming	Quick
Filtering	Not available	Available
Cognitive Load	Higher	Reduced

9.3 Trade-offs

The Python-based visualization approach is relatively simple to implement, test, and modify. Interactive dashboards also provide better flexibility for exploring the dataset.

However, interactive dashboards require additional development effort compared to static charts. The effectiveness of the analysis also depends on the size and quality of the available dataset.

9.4 Justification of the Final Solution

The proposed solution is suitable for demonstrating Data Handling and Visualization concepts. It provides a clear process from raw data to decision-ready visualizations.

The interactive dashboard allows users to explore customer ordering patterns and compare different categories. The solution is therefore appropriate for the academic objective of analyzing data patterns, evaluating visualization designs, and creating effective interactive visualizations.

10. SDG Relevance

The project is related to United Nations Sustainable Development Goal SDG 8: Decent Work and Economic Growth. Data-driven analysis can improve business efficiency and help organizations make better operational and marketing decisions.

The project is also related to SDG 12: Responsible Consumption and Production. Understanding food ordering patterns can help businesses plan inventory and demand more effectively, potentially reducing unnecessary food preparation and wastage.

The SDG connection is considered as a broader relevance of the project.

11. Conclusion, Limitations and Possible Improvements

11.1 Conclusion

The project successfully demonstrates the process of transforming raw food delivery customer data into meaningful visual insights and an interactive analytics dashboard. The implementation applies important Data Handling and Visualization concepts, including data cleaning, data wrangling, trend analysis, comparative visualization, correlation analysis, visualization evaluation, and interactive dashboard development.

The successful processing of the dataset confirms that the proposed analysis pipeline can identify useful patterns and present them in a clear and understandable form.

11.2 Limitations

The current implementation has some limitations:

- The dataset contains only 200 records.
- The analysis is limited to the available variables.
- Real-time data is not included.
- The dashboard focuses mainly on descriptive analysis.
- The dataset does not provide detailed product-level information for advanced market-basket analysis.
- Customer demographic information is limited.

11.3 Possible Improvements

Future versions could improve the system by:

1. Using larger datasets.
2. Adding real-time food delivery data.
3. Performing customer segmentation.
4. Adding sales prediction and forecasting.
5. Implementing recommendation systems.
6. Adding more interactive dashboard features.
7. Analyzing customer retention patterns.
8. Performing advanced association analysis.
9. Including more detailed performance metrics.

12. Individual Contribution of Group Members

The project work was divided among the group members. Each member contributed to a specific part of data analysis, visualization, dashboard development, testing, and documentation.

Group Member	Contribution
A Roja	Worked on data loading, data cleaning, and data wrangling.
M Arshini	Worked on exploratory data analysis and visualization generation.

Group Member	Contribution
Akshaya	Worked on dashboard development, testing, result analysis, and report preparation.
All Group Members	Participated in project discussions, testing, visualization evaluation, and final report preparation.

13. References

- [1] Python Software Foundation, “Python 3 Documentation,” Python Software Foundation. [Online]. Available: Python Documentation
- [2] The pandas development team, “pandas Documentation,” pandas. [Online]. Available: pandas Documentation
- [3] W. S. Cleveland and R. McGill, “Graphical perception: Theory, experimentation, and application to the development of graphical methods,” *Journal of the American Statistical Association*, vol. 79, no. 387, pp. 531–554, 1984.
- [4] E. R. Tufte, *The Visual Display of Quantitative Information*, 2nd ed. Cheshire, CT, USA: Graphics Press, 2001.
- [5] United Nations, “Sustainable Development Goals,” United Nations. [Online]. Available: United Nations Sustainable Development Goals

14. One-Page Individual Reflection

I started by loading the food delivery customer dataset using Python and Pandas. I examined the structure of the dataset, the available columns, data types, and number of records. I also checked the dataset for missing values and duplicate records. This process helped ensure that the data used for further analysis was reliable and consistent.

Another important task was processing the **Order_Date** column and converting it into a suitable datetime format. This made the dataset ready for time-based analysis and helped support the identification of order trends. I also organized and prepared the required variables so that the dataset could be effectively used for further analysis and visualization.

One of the major things I learned from this project was that data cleaning and data wrangling are essential steps in data analysis. If the underlying data is incorrect or poorly structured, even a well-designed visualization may produce misleading results. I gained practical experience using Pandas functions to inspect, clean, transform, and organize the dataset.

Overall, this assignment helped me achieve the course outcomes **CO3, CO4, and CO5**. I gained practical knowledge of Python and Pandas and developed a better understanding of how raw data is converted into an analysis-ready dataset. The project improved my data handling skills and gave me valuable practical experience in the initial stages of a complete data analytics and visualization process.